An empirical example: *Rhinoclemmys* data

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General description

I conducted a phylogenetic diversity (PD) analysis for the *Rhinoclemmys* genus to evaluate the effect of branch length on the chosen conservation area(s). The topology used corresponds to a total evidence analysis from (Romero-Alarcon 2020), while the distribution was modified from (Le and McCord 2008).

Read the data: the distribution, and the tree

```
## For reproductibility purposes

set.seed(121)


options(warn=-1)
library(ggplot2)
suppressMessages(library(ggtree))

suppressMessages(library(blepd))

suppressMessages(library(phytools))

## library(ggtree)

###
## If you want to install ggtree.
##
## https://bioconductor.org/packages/release/bioc/html/ggtree.html
##
####
##  if (!require("BiocManager", quietly = TRUE))
##      install.packages("BiocManager")
##  BiocManager::install("ggtree")
###

###
## Version
###

cat("Analyses made with blepd version:",
    unlist(packageVersion("blepd")),"\n")
```

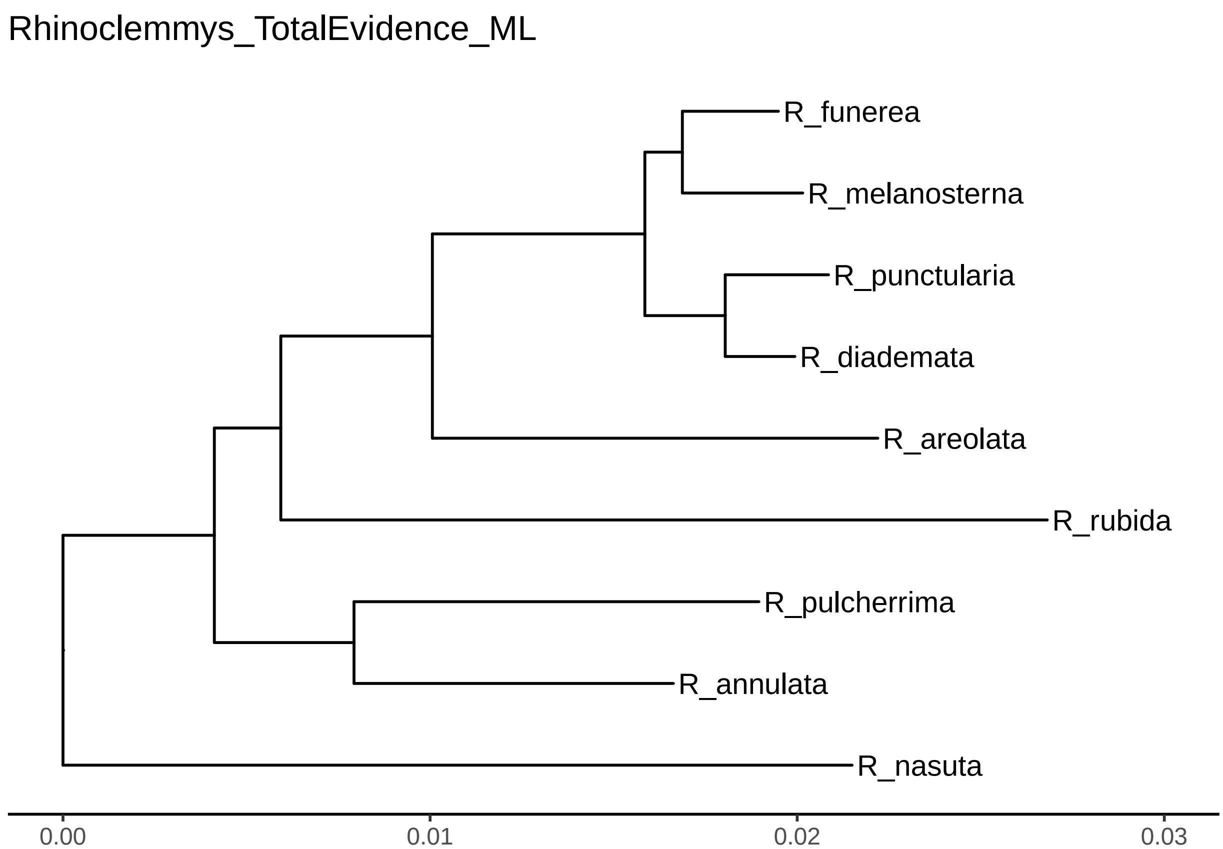
```
## Analyses made with blepd version: 0 3 8 2025 1 10
```

```r
###
## The tree file (in Newick format) and the distribution file (in CSV format)
## can be found in the data directory or as a data set within the package.
###

###
## Assuming data is read from files, create an object to store the
## distribution and the tree.
###

RhinoclemmysData <- list()

## Read data

## The distribution is a labeled CSV file with areas as rows and terminals as
## columns.

#setwd("./csv/")

## getwd()

csvFile <- list.files(pattern="csv")

## csvFile

### The functions use a matrix object for the distributions

RhinoclemmysData$distribution <-  as.matrix(read.table(csvFile,
                                   stringsAsFactors=FALSE,
                                   header=TRUE,
                                   row.names=1,
                                   sep=",")
                                   )


print(t(RhinoclemmysData$distribution))
```

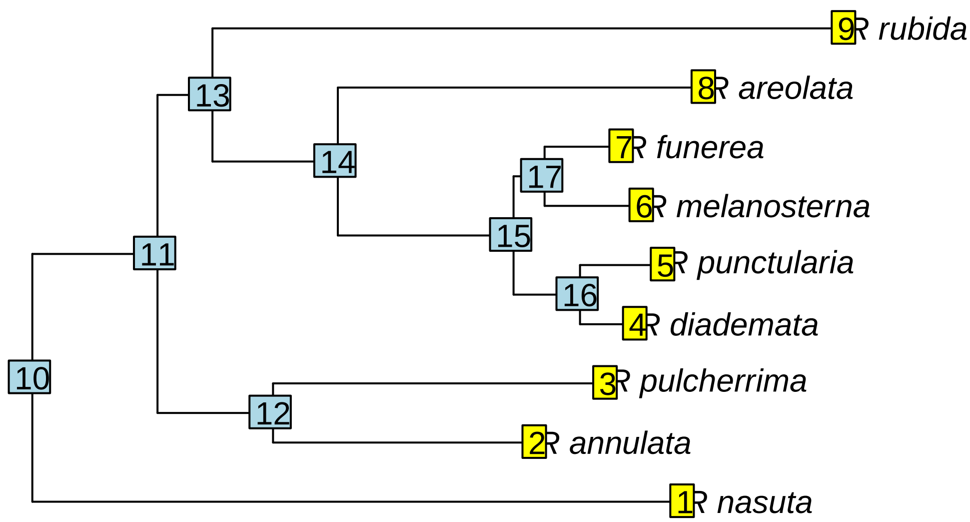
```
##                  Al An-As Cho Nsa Nuh Ph Pl
## R_annulata        1     0   1   0   0  1  0
## R_areolata        1     0   0   0   0  0  0
## R_diademata       0     0   0   1   0  0  0
## R_funerea         1     0   0   0   1  0  0
## R_melanosterna    0     0   1   0   0  1  0
## R_nasuta          0     0   1   0   0  0  0
## R_pulcherrima     0     0   0   0   1  0  1
## R_punctularia     0     1   0   0   0  0  0
## R_rubida          0     0   0   0   0  0  1
```

```r
###
## Tree(s) in nexus or newick format
##
## setwd("../tree/")
###
```

```r
treeFiles <- list.files(pattern=".tre")

##treeFiles

RhinoclemmysData$tree  <-    read.tree(treeFiles)


## name of tree(s) without .tre

treeFiles <- gsub(".tre","",treeFiles)


##getwd()

###
## You might need to save the object as an .Rda file.
##
## save(RhinoclemmysData, file = "../data/RhinoclemmysData.rda")
###


## Plotting:

## The tree

## using ggtree

# RhinoclemmysData$tree <- reorder(RhinoclemmysData$tree, order = "cladewise")

plotTree <-  ggtree(RhinoclemmysData$tree, ladderize=TRUE,
                    color="black", size=0.51, linetype="solid") +
             geom_tiplab(size=4, color="black") +
             xlim(0,0.030) +
             theme_tree2() +
             ggtitle(treeFiles[1])


##pdf("RhinoclemmysTopology.pdf", onefile = TRUE)

print(plotTree)
```




```r
##dev.off()

###
## Alternatively, we can plot the tree using APE
###

plot.phylo(RhinoclemmysData$tree)

###
## The number of the nodes / tips
##
nodelabels()
#
tiplabels()
```




```
###
```

#Plot the branch lengths.

```
library(ggplot2)

##par(mfrow=c(2,1))

datos <- RhinoclemmysData$tree$edge.length


cat("s.d. All=",formatC(sd(datos),format = "e", digits = 1),"\n")
```

```
## s.d. All= 6.5e-03
```

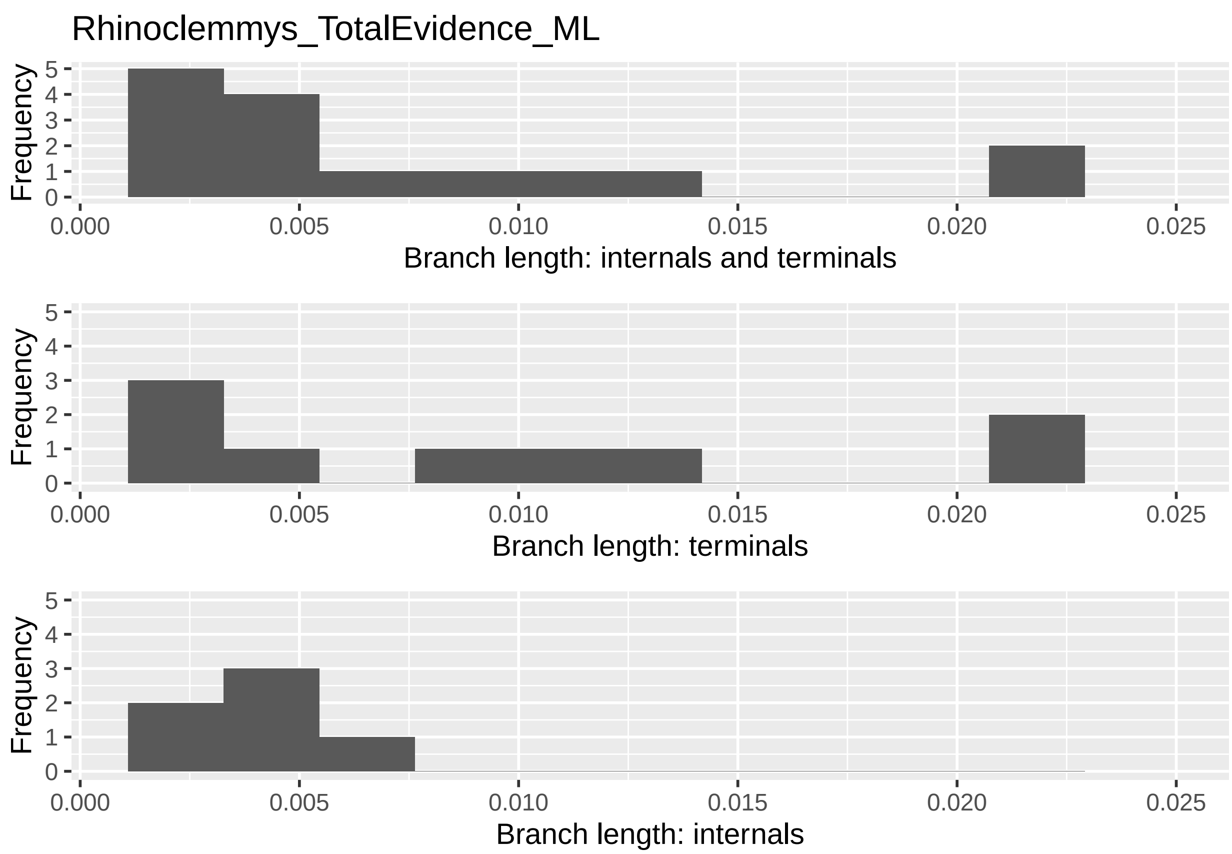
```
data <- as.data.frame(datos)

a <- ggplot2::ggplot(data, aes(x=datos)) +
     geom_histogram(bins = 12) +
     labs(title = treeFiles, x = "Branch length: internals and terminals", y = "Frequency") +
     xlim(c(0.001, 0.025)) +
     ylim(c(0, 5))

terminals <- RhinoclemmysData$tree$edge[,2] < 1 + length(RhinoclemmysData$tree$tip.label)

datos <- RhinoclemmysData$tree$edge.length[terminals]

cat("s.d. Terminals=",formatC(sd(datos),format = "e", digits = 1),"\n")
```

```
## s.d. Terminals= 7.7e-03
```

```r
data <- as.data.frame(datos)

b <- ggplot(data, aes(x=datos)) +
  geom_histogram(bins = 12) +
  labs(x = "Branch length: terminals", y = "Frequency") +
  xlim(c(0.001, 0.025)) +
  ylim(c(0, 5))



datos <- RhinoclemmysData$tree$edge.length[!terminals]


cat("s.d. Internals=",formatC(sd(datos),format = "e", digits = 1),"\n")
```

```
## s.d. Internals= 1.7e-03
```

```r
data <- as.data.frame(datos)

c <- ggplot(data, aes(x=datos)) +
  geom_histogram(bins = 12) +
  labs(x = "Branch length: internals", y = "Frequency") +
  xlim(c(0.001, 0.025)) +
  ylim(c(0, 5))



##pdf("RhinoclemmysBranchLength.pdf", onefile = TRUE)

cowplot::plot_grid(a, b, c, nrow=3)
```




```
##dev.off()

###
## cat("Terminals, with BL larger than 0.02 = ",
## findTerminalgivenLength(RhinoclemmysData$tree,0.02))
###
```

The branch length histograms and the tree plot showed that the internal length branches were similar among each other, and different to terminals'; there were two longer branches (larger than 0.02), *R. nasuta* (inhabiting Cho) and *R. rubida* (Pl), whereas Al and Cho were the richest.

PD values

First, I calculated the PD value for the areas, given this tree.

```
RhinoclemmysData$tablePD <- PDindex(RhinoclemmysData$tree,
                              distribution=RhinoclemmysData$distribution,
                                    root=TRUE,
                                    percentual = TRUE)

RhinoclemmysData$matrixPD <- as.data.frame(
                                  matrix(
                                      unlist(RhinoclemmysData$tablePD),
                                      nrow= length(treeFiles),
                                      byrow=TRUE))
```

```
colnames(RhinoclemmysData$matrixPD) <- row.names(RhinoclemmysData$distribution)

row.names(RhinoclemmysData$matrixPD) <- "PD value"

 new <- (t(RhinoclemmysData$matrixPD))


values <-  new[order(new[,1])]
names  <-  rownames(new)[order(new[,1])]

print(as.data.frame(paste(names,values)))
```

```
##   paste(names, values)
## 1             Nsa 8.05
## 2           An-As 8.42
## 3             Ph 13.18
## 4            Nuh 13.86
## 5             Pl 16.81
## 6             Al 17.82
## 7            Cho 21.86
```

The highest PD was for area Cho, followed by Al, the two richest areas. However, the difference in PD values was [?] not determined by species richness alone, but also by the branch lengths of the species within each region.

Effect of branch lengths in the PD

I first investigated the impact of the number of replicates on the results.

```
for(repetir in 1:4){

            val <- swapBL( RhinoclemmysData$tree,
                           RhinoclemmysData$distribution,
                           model = "allswap",
                           branch = "all",
                           nTimes = 10**repetir
                         )

            printSwapBL(val)

}
```

```
## model to test allswap reps 10
##   Initial Branch   Model Al Cho Nuh
## 1     Cho    all allswap 20  70  10
## model to test allswap reps 100
##   Initial Branch   Model Al An-As Cho Nsa Nuh Pl
## 1     Cho    all allswap 30     3  42   3  19  3
## model to test allswap reps 1000
##   Initial Branch   Model   Al An-As  Cho Nsa  Nuh Pl
## 1     Cho    all allswap 35.6   4.1 40.1   4 14.2  2
## model to test allswap reps 10000
##   Initial Branch   Model   Al An-As   Cho  Nsa   Nuh   Pl
## 1     Cho    all allswap 36.7  4.22 38.81 4.48 13.94 1.85
```

I found that results stabilized after approximately 1000 replicates, with most differences arising from using 10 or 100 replicates. As a general guideline, at least 100 replicates are necessary, but 1000 are preferable.

I tested the effect of branch length on PD values by swapping terminal and/or internal branch lengths using three different models.

```
for( modelo in c("simpleswap","allswap","uniform") ){

       for( rama in c("terminals","internals","all") ){

           val <- swapBL(RhinoclemmysData$tree,
                         RhinoclemmysData$distribution,
                         model   =  modelo,
                         branch  =  rama,
                         nTimes  =  1000
              )

        cat( "\n\t Tree=",treeFiles,
              "\n\t Model=",modelo,
              "\n\t Branchs swapped=",rama,"\n" )

       printSwapBL(val)

        cat( "\n\n" )

    }

 }
```

```
## model to test simpleswap reps 1000
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= simpleswap
##   Branchs swapped= terminals
##   Initial    Branch      Model   Al Cho  Pl
## 1     Cho terminals simpleswap 15.4  81 3.6
##
##
## model to test simpleswap reps 1000
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= simpleswap
##   Branchs swapped= internals
##   Initial    Branch      Model Cho
## 1     Cho internals simpleswap 100
##
##
## model to test simpleswap reps 1000
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= simpleswap
##   Branchs swapped= all
##   Initial Branch      Model   Al  Cho  Pl
## 1     Cho    all simpleswap 11.1 87.8 1.1
##
```

```
##
## model to test allswap reps 1000
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= allswap
##   Branchs swapped= terminals
##   Initial    Branch   Model   Al An-As  Cho Nsa  Nuh  Pl
## 1     Cho terminals allswap 37.2   1.4 45.2 1.6 11.4 3.2
##
##
## model to test allswap reps 1000
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= allswap
##   Branchs swapped= internals
##   Initial    Branch   Model Cho
## 1     Cho internals allswap 100
##
##
## model to test allswap reps 1000
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= allswap
##   Branchs swapped= all
##   Initial Branch   Model   Al An-As  Cho Nsa  Nuh  Pl
## 1     Cho    all allswap 36.2   4.4 40.9 4.7 12.2 1.6
##
##
## model to test uniform reps 1000
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= uniform
##   Branchs swapped= terminals
##   Initial    Branch   Model   Al An-As  Cho Nuh  Pl
## 1     Cho terminals uniform 48.5   0.1 46.4 4.6 0.4
##
##
## model to test uniform reps 1000
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= uniform
##   Branchs swapped= internals
##   Initial    Branch   Model Cho
## 1     Cho internals uniform 100
##
##
## model to test uniform reps 1000
##
##   Tree= Rhinoclemmys_TotalEvidence_ML
##   Model= uniform
##   Branchs swapped= all
##   Initial Branch   Model   Al  Cho Nsa Nuh
## 1     Cho    all uniform 46.3 47.5 0.1 6.1
```

I determined that internal branch length has no impact on the swapping evaluation, while the terminal branch lengths were crucial in the chosen area. For most models tested, Cho was the selected area. Only when I swapped all or terminal branch lengths (model = "allswap") or replaced them with a uniform distribution (model = "uniform") did the optimal area shift from Cho to Al, but at a "significant" level.

Recognizing the uneven distribution of terminal branch lengths, I suspected that the longest branches within the Cho and/or Al areas were driving these results. Therefore, I investigated whether the results could be attributed to the lengths of the longest branches.

```
AllTerminals <- evalBranch(tree=RhinoclemmysData$tree,
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="terminals",
                           approach="all",
                           printNames=TRUE)

printEvalBranch(AllTerminals)
```

```
##               node initialArea FinalArea Aproach %Delta
## 1       [R_nasuta]         Cho        Al   lower    -46
## 12 [R_pulcherrima]         Cho        Pl   upper    116
## 13   [R_diademata]         Cho       Nsa   upper   1710
## 14 [R_punctularia]         Cho     An-As   upper   1104
## 16     [R_funerea]         Cho        Al   upper    326
## 17    [R_areolata]         Cho        Al   upper     86
## 18      [R_rubida]         Cho        Pl   upper     59
```

```
AllInternals <- evalBranch(tree=RhinoclemmysData$tree,
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="internals",
                           approach="all",
                           printNames=TRUE)

printEvalBranch(AllInternals)
```

```
##                            node initialArea FinalArea Aproach %Delta
## 13 [R_diademata/R_punctularia]         Cho     An-As   upper   1695
```

I found that terminal branch length influenced the selected area. While modifying internal branches had minimal to no effect, altering terminal branch lengths could impact area selection. Specifically, a 46% decrease in the branch length of *R. nasuta* qor an 86% increase in the branch length of *R. areolata* shifted the optimal area from Cho to Al. Similarly, a 59% increase in the branch length of *R. rubida* or a 116% increase in the branch length of *R. pulcherrima* resulted in a shift from Cho to Pl.

These findings suggest that decisions to conserve solely area Cho are misguided. Areas Al, and potentially Pl, should be considered essential components of the conservation range.

Literature cited

Le, Minh, and William P. McCord. 2008. "Phylogenetic relationships and biogeographical history of the genus *Rhinoclemmys* Fitzinger, 1835 and the monophyly of the turtle family Geoemydidae (Testudines: Testudinoidea)." *Zoological Journal of the Linnean Society* 153 (4): 751–67. https://doi.org/10.1111/j.1096-3642.2008.00413.x.

Romero-Alarcon, L. Viviana. 2020. "A total-evidence phylogeny of the crown and stem-groups of turtles (Pantestudines: Testudinata)." Master's thesis, Colombia: Escuela de Biología, UIS.